

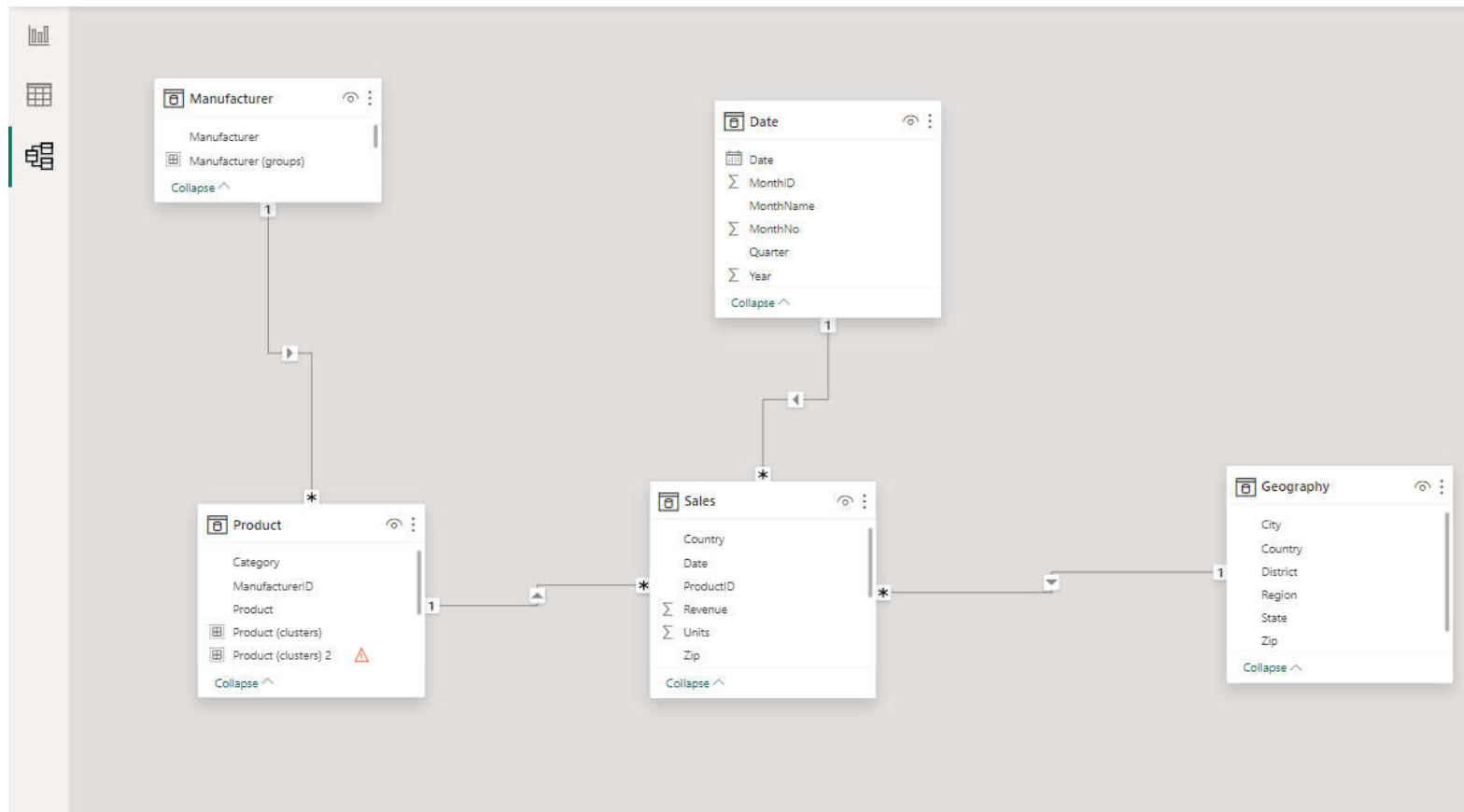
Microsoft Power BI

Part 2

Report



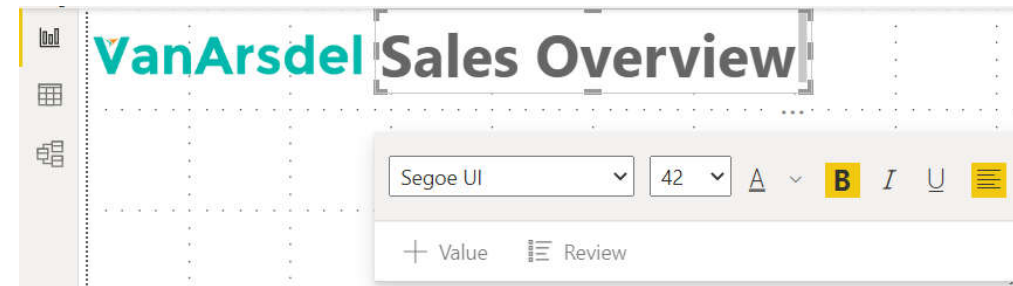
Data Model-Model View



Power BI Desktop – Add Image and Text

You need the company logo in most reports.

1. Click on the white space on the canvas.
2. From the ribbon, select Insert -> Image.
3. Navigate to image file and select Logo.png. Click Open.
4. Logo will appear on the report.
5. Resize and move the logo to the top left of the report.



Next let us add a title to this report. To do so click anywhere on the white canvas.

1. From the ribbon, select Insert -> Text box. Text box area is available in the canvas.
2. Enter the title "Sales Overview".
3. Highlight Sales section of the header and change the font to Segoe UI, font size to 42 and make the text bold.

Power BI Desktop – Line graph

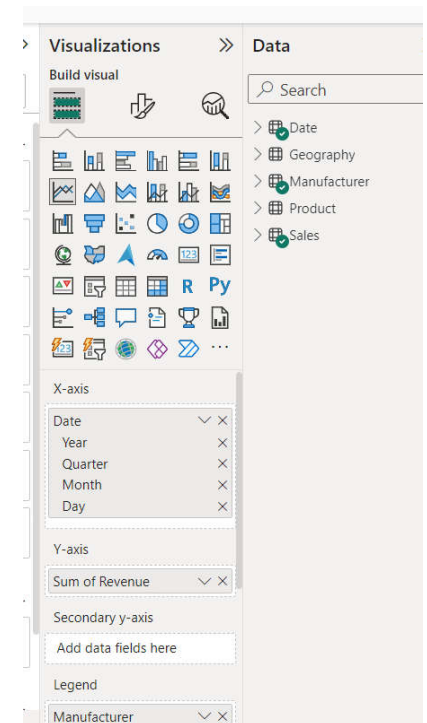
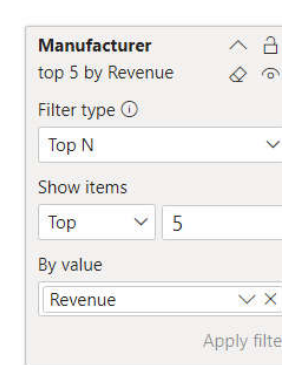
1. From Visualizations section select Line Chart .
2. From the Data section expand Sales table and drag and drop Revenue to Y-axis.
3. From the Data section expand Date table and drag and drop Date to X-axis
4. From the Data section, expand Manufacturer table and drag and drop Manufacturer to the Legend of the visual
5. From Filters section, scroll to Visual level filters.
6. Expand Manufacturer and from the Filter Type drop down select Top N.
7. Enter 5 next to Top since we are comparing the Top 4 competitors
8. From the Data section, expand Sales table and drag and drop Revenue to By Value.

We are doing this to get the Top 5 competitors based on revenue.

9. Select Apply Filter

It will be useful to display Units sold along with Revenue information on the tooltip.

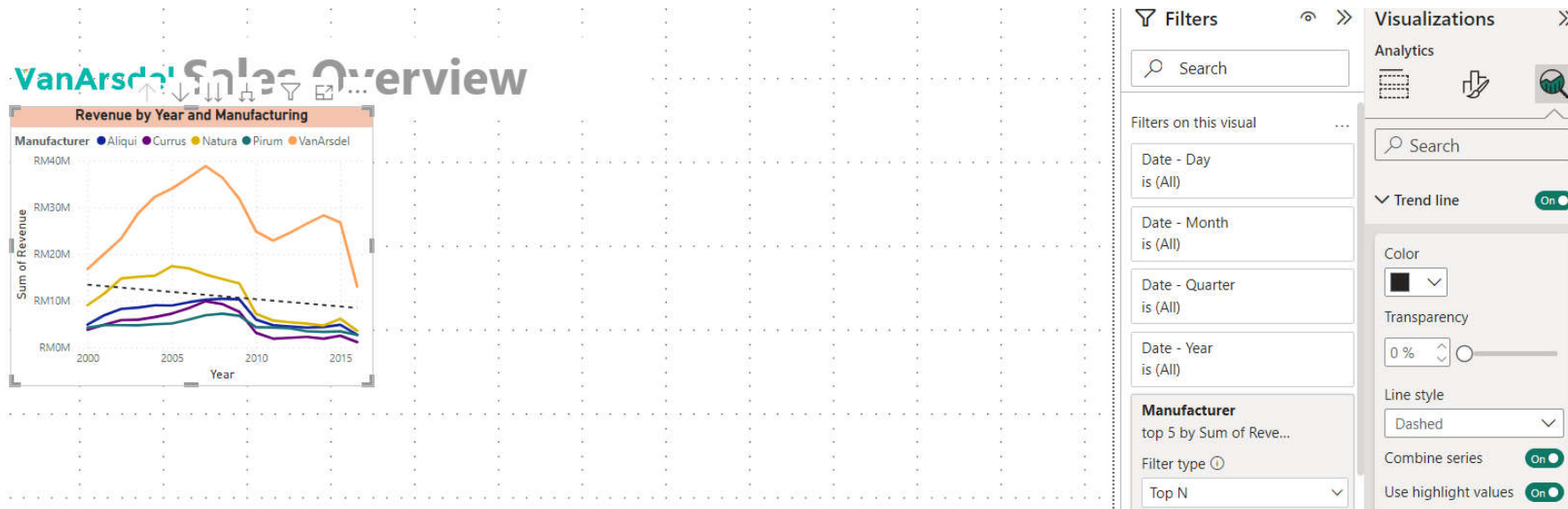
10. From the Data section, expand Sales and drag Units to Tooltips



Power BI Desktop – Trend Line

Power BI Desktop provides capability to perform analytics on visuals. It's useful to know the revenue trend. Let's add it to the Line chart.

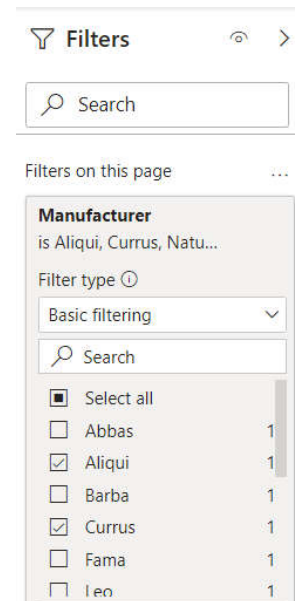
1. With the Line chart selected, from the Visualizations click on the Analytics icon.
2. Notice there is options to add Trend, Constant, Min, Max, Average, Median and Percentile Line.
3. Turn On Trend line



Power BI Desktop – Trend Line

Earlier we used Top N feature to filter the column chart down to Top competitors. Suppose we want to apply this filter to all the visuals in a page,

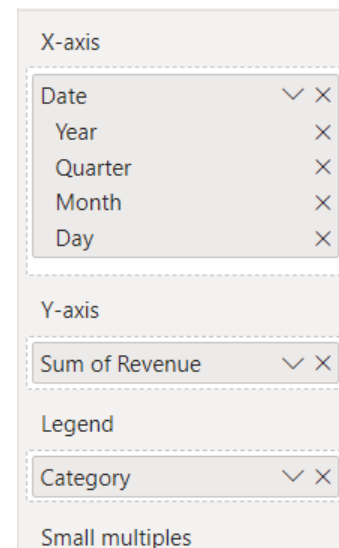
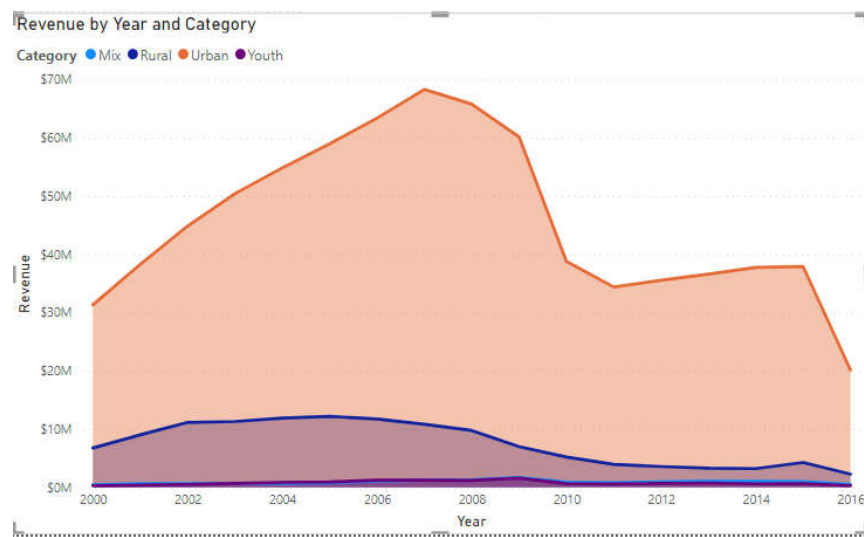
1. Click on the white space on the canvas
2. In Filters section , add the Manufacturer column to Filters on this page
3. Select the manufacturers Aliqui, Currus, Natura, Pirum and VanArsdel in the filter



Power BI Desktop – Area Graph

Revenue by Category

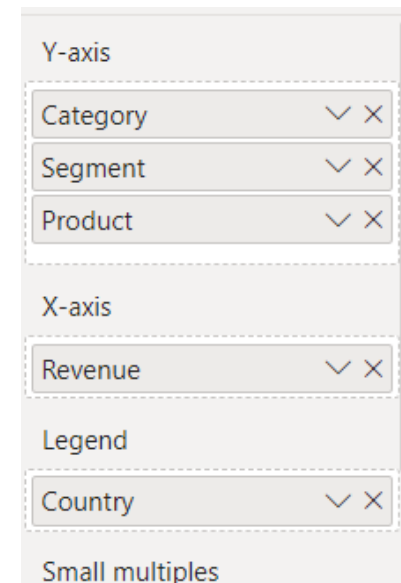
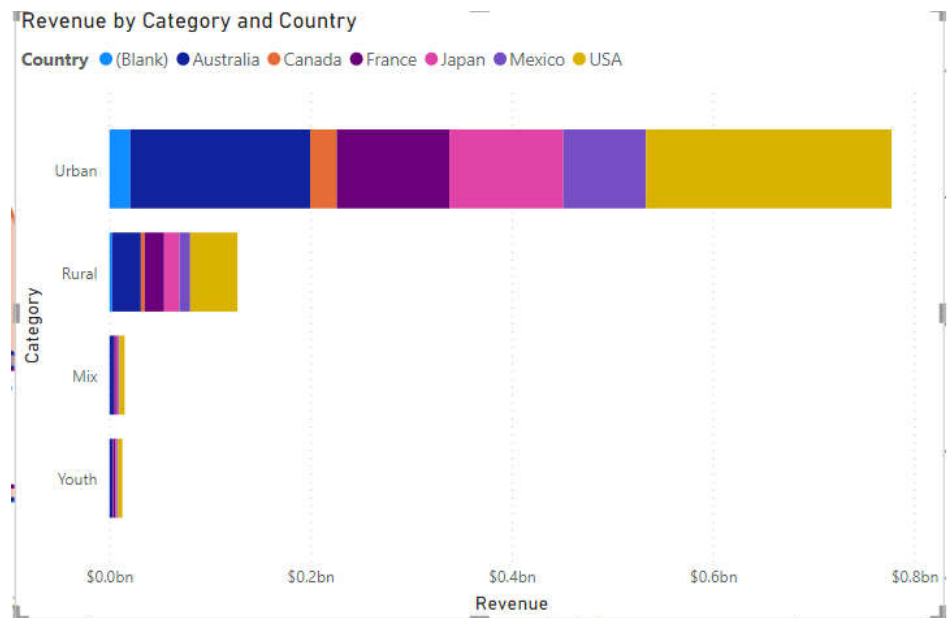
1. Click in the report canvas.
2. Select the Stacked area chart visual from the Visualizations section.
3. From the Data section, expand Date table and drag and drop Date field to X-axis.
4. From the Data section, expand Product table and drag and drop Category field to Legend.
5. From the Data section, expand Sales table and drag and drop Revenue field to Y-axis.
6. Resize the chart as needed.



Power BI Desktop – Hierarchy Stack Bar Chart

This chart provides Revenue by Country by Product hierarchy

1. Select the Stacked bar chart visual from the Visualizations section.
2. From the Data section, expand Product table and drag and drop Category ,Segment and Product field to Y-axis
3. From the Data section, expand Geography table and drag and drop Country field to Legend.
4. From the Data section, expand Sales table and drag and drop Revenue field to X-axis.
5. Feel free to drill up and down the Product hierarchy and analyze the revenue



DAX (Data Analysis Expressions)

- DAX (Data Analysis Expressions) is a formula expression language include functions, operators, and values to perform advanced calculations and queries on data in related tables and columns in tabular data models.
- By using Data Analysis Expressions (DAX), you can add three types of calculations to your data model:
 - i. Calculated tables
 - ii. Calculated columns
 - iii. Measures

Power BI Desktop – DAX (Year To Date)

You want the ability to compare current Year to Date sales with the previous year and see if you are doing better or worse. To achieve such calculations in Power BI desktop you need to create a Measure using a DAX functions that understand Time calculations such as “Year to Date”

- 1.To create a new Measure, from Data section click on the ellipsis next to Sales table.
2. Select New Measure. Notice a formula bar appears.
3. In the formula bar enter the following DAX Expression

```
YTD Revenue = TOTALYTD(sum(Sales[Revenue]),'Date'[Date])
```

- 4.Click on the check mark next to the formula bar.
- 5.You will see the YTD Revenue under Sales table with a special icon which indicates it is a measure (calculator).

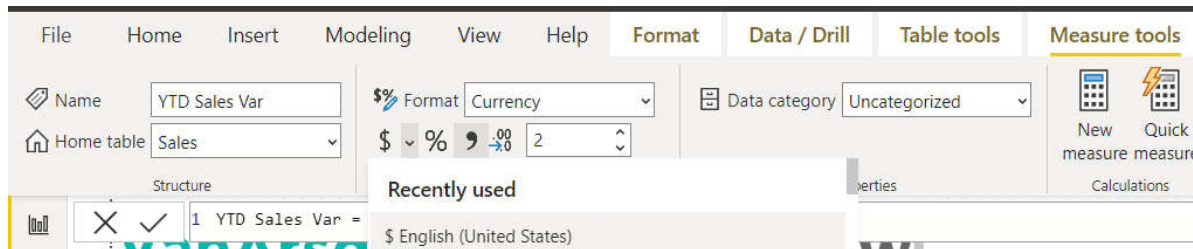
Similarly, let's create Last Year to Date Revenue measure and Year To date Sales variable

```
LY YTD Revenue = CALCULATE([YTD Revenue],SAMEPERIODLASTYEAR('Date'[Date]))
```

```
YTD Sales Var = [YTD Revenue]-[LY YTD Revenue]
```

Power BI Desktop – Formatting

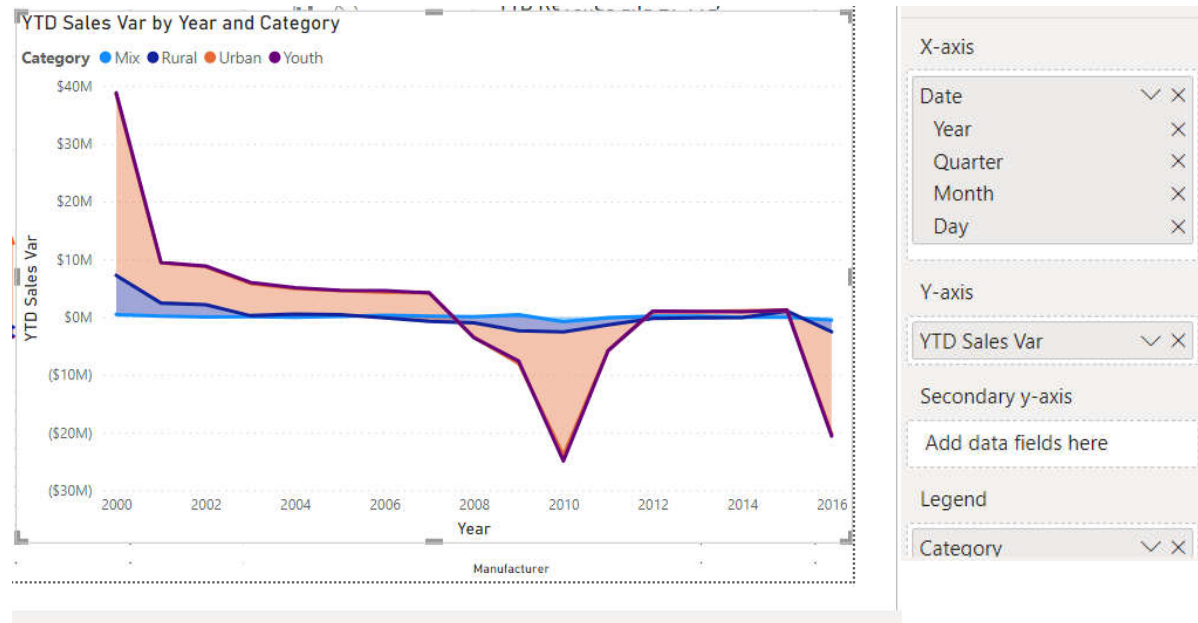
- Let's format the newly created measures.
- From the Data section, expand Sales table and select YTD Revenue
 - From the ribbon, select Measure Tools -> Format -> Currency -> \$ English (United States).
 - Similarly, format LY YTD Revenue and YTD Sales Var measures



Power BI Desktop

This chart provides sales trend by product category over time

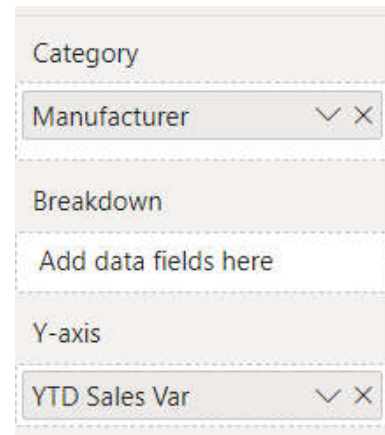
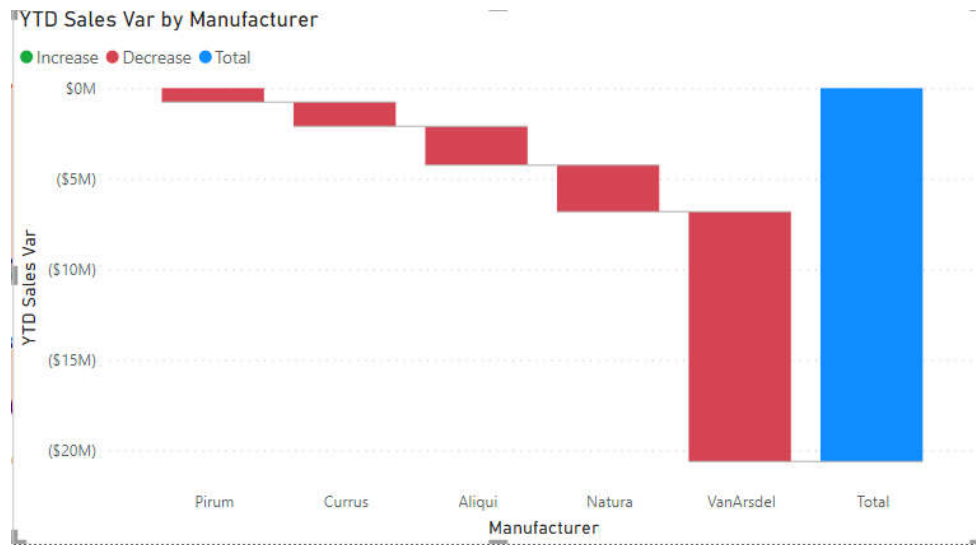
1. Select the Stacked bar chart visual from the Visualizations section.
2. From the Data section, expand Date table and drag and drop date field to X-Axis.
3. From the Data section, expand Product table and drag and drop Category field to Legend.
4. From the Data section, expand Sales table and drag and drop YTD Sales Var field to Y-Axis



Power BI Desktop-Waterfall chart

This chart provides the analysis to find out how all the Manufacturers have performed overtime.

1. Select the Waterfall chart visual from Visualizations.
2. From the Data section, expand Manufacturer table and drag and drop Manufacturer field to Category section.
3. From the Data section, expand Sales table and drag and drop YTD Sales Var field to Y-axis

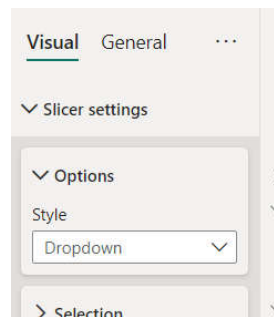


Power BI Desktop-Slicer

A slicer provides the ability for end users to visually see the key filters applied to the entire page and the flexibility to select/deselect values easily.

Date slicer

1. Click on the slicer icon in Visualizations pane.
2. From the Data section, expand Date table and select Year column and drag and drop to Field section
3. Go to Format and select Slicer Settings . Select Dropdown to change the slicer from a List to a Dropdown. Resize the slicer as needed.
4. Move the slicer to the top right corner of the canvas.



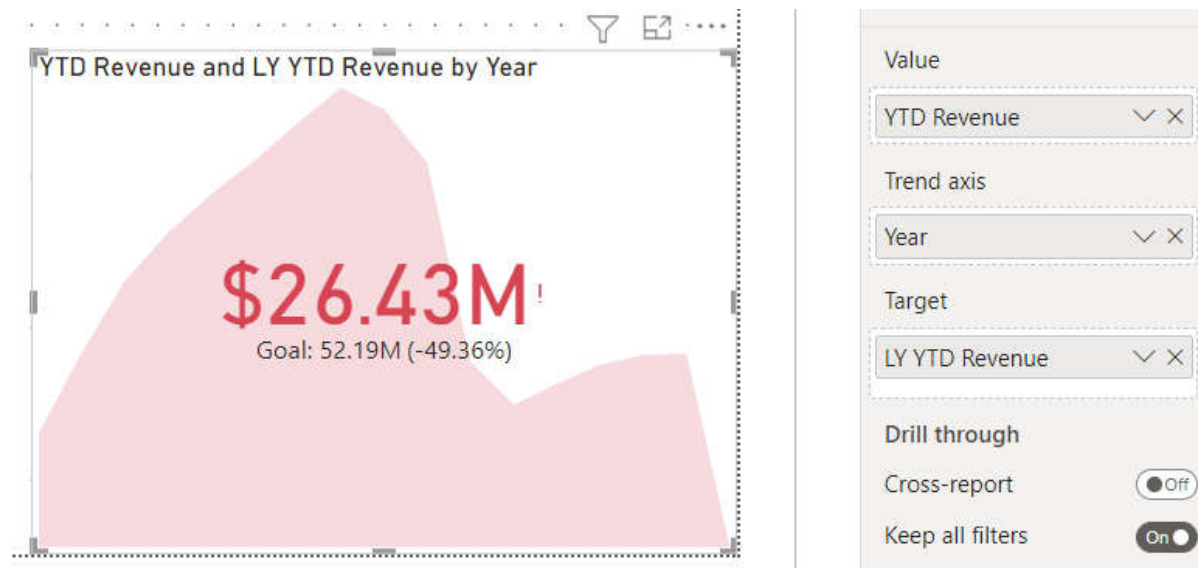
Let's add a slicer which will help analyze the data by Product Segment.

1. Click on the slicer icon in Visualizations pane.
2. From Data section, expand Product table and select Segment column and drag and drop to Field section .
3. Change the slicer from a List to a Dropdown.

Power BI Desktop-KPI

Let's add a KPI visual that compares this year's revenue versus last year revenue

1. Click on the whitespace in the canvas.
2. Click on the KPI visual in Visualizations section.
3. From the Data section, expand Date table and drag and drop Year field to Trend axis section.
4. From the Data section, expand Sales table and drag and drop YTD Revenue field to value section.
5. From the Data section, expand Sales table and drag and drop LY YTD Revenue field to Target section

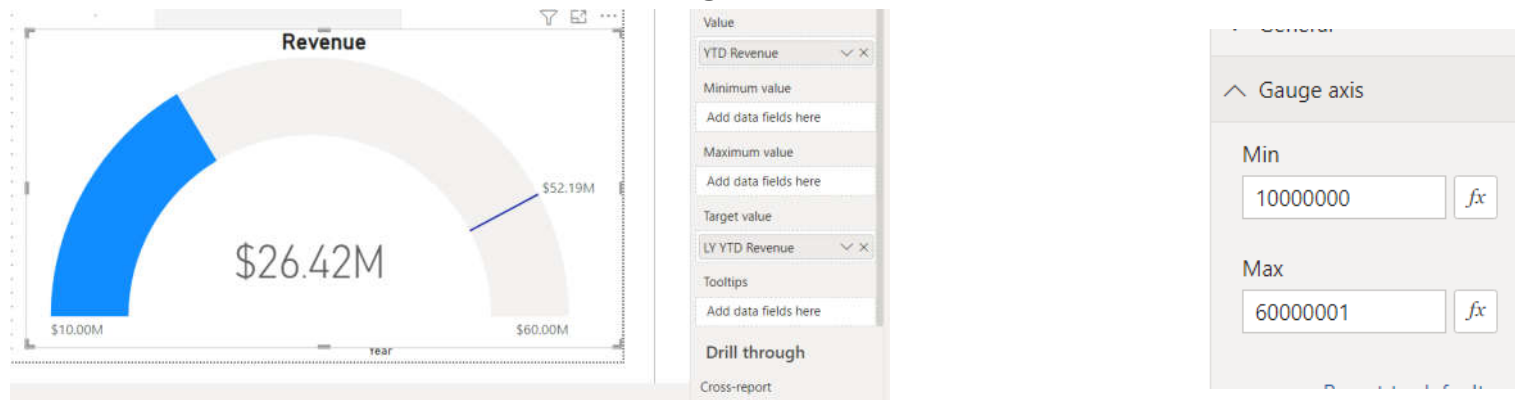


Power BI Desktop-Gauge Card

- A radial gauge chart has a circular arc and shows a single value that measures progress toward a goal or a Key Performance Indicator (KPI).
- The line (or needle) represents the goal or target value.
- The shading represents the progress toward that goal.
- The value inside the arc represents the progress value. Power BI spreads all possible values evenly along the arc, from the minimum (left-most value) to the maximum (right-most value)

Revenue Performance

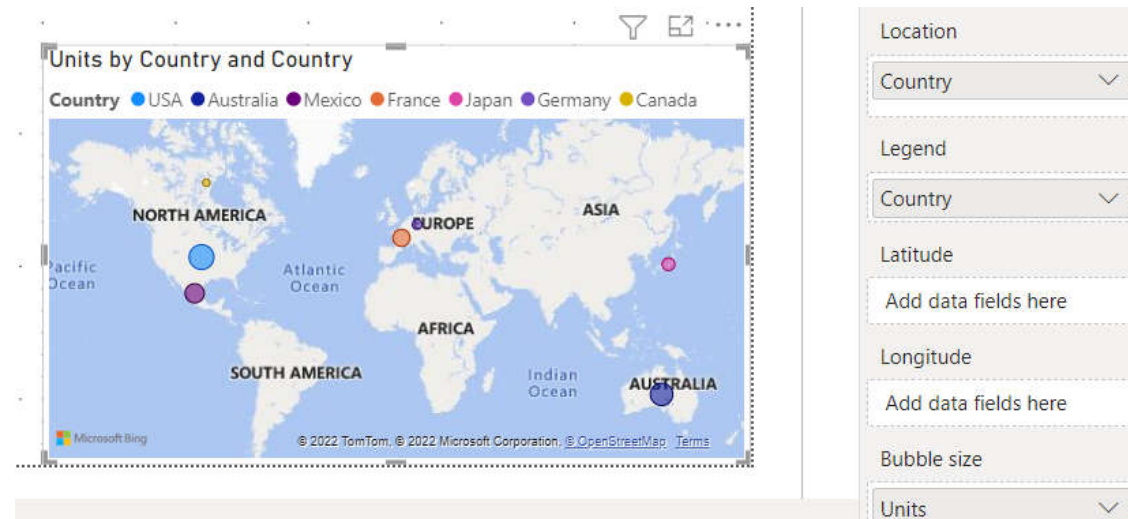
1. From Visualizations section select Gauge Card chart.
2. From the Data section expand Sales table and drag and drop YTD Revenue to value and LY YTD revenue to Target value



5. With the gauge card chart selected, from the Visualization click on the Format icon. Modify Min and Max value

Power BI Desktop- Map

1. Click in the report canvas. Select the map visual from Visualizations.
2. From Data section, expand Geography table and drag and drop Country column to Location section of the map visual
3. From Data section, expand Sales table and drag and drop Units column to BubbleSize section of the visual.
4. From Data section, expand Geography table and drag and drop Country field to Legend section
5. Resize the map as needed. The map visual provides units sold by country



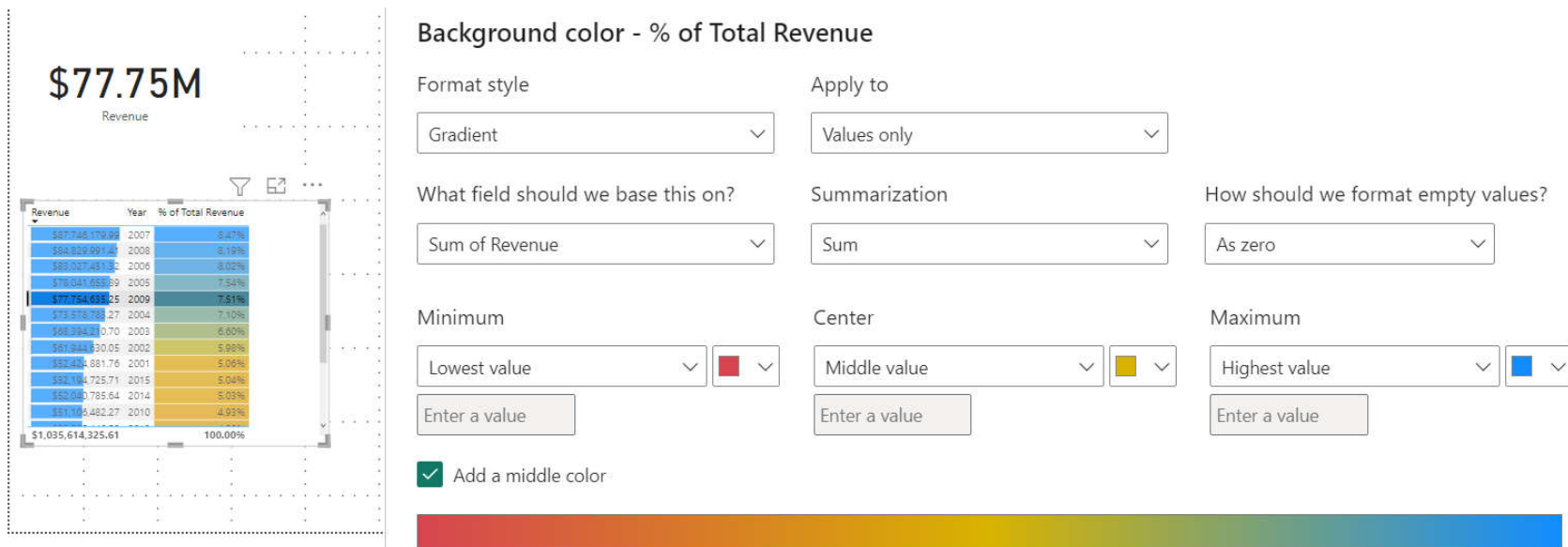
Power BI Desktop-Table

Let's assume we want to create a table visual to display the Revenue by year and add a column that displays the % of revenue for each year to help figure out the years where sales were high

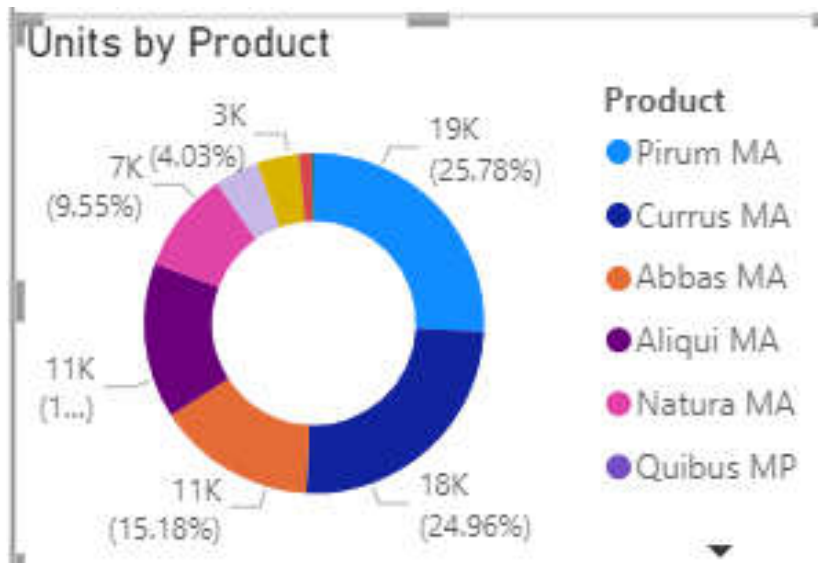
1. From the Visualizations section, select Table.
2. From the Data section, and select Year. expand Date table
3. From the data section, expand Sales and select Revenue.
Notice that Year is summed by default since the data type is of type Whole Number. We would never sum Year, so let's change it.
4. In the Data section, click on Year.
5. From the ribbon, select Column Tools -> Summarization -> Do Not Summarize.
6. From the column section of the Table visual, delete Sum of Year field by clicking on the "x".
7. From the Date table, drag Year field and drop it above Revenue in Column section of the table visual.
8. From the Column section, select the arrow next to Revenue field.
9. Select Conditional formatting -> Data bars. Data bars dialog opens. There are options to format the data bar. Select OK. Notice data bar helps to easily identify the years with high revenue and those years with low revenue.

Power BI Desktop-Table(cont.)

- 10 From the Sales table, drag Revenue field and drop it below Revenue in column section of the table visual..
11. Click the arrow next to the second Revenue field. Select Show value as -> Percent of grand total.
15. In the Column Section of the table report, click on the arrow next to % GT Revenue.. From the menu select Rename. Rename the field to % of Total Revenue.



Donut Chart



Visualizations >>

Build visual

Donut chart

Legend

Product

Values

Units

Details

Card

Revenue	Units sold
\$26.21M	72K

Visualizations >>

Build visual

Card

Fields

Units

Drill through

Cross-report ☐ Off

Keep all filters ☒ On

Add drill-through fields here

Visualizations >> D

Build visual

Fields

Revenue

Drill through

Cross-report ☐ Off

Keep all filters ☒ On

Add drill-through fields here

Power BI Desktop-Smart Narrative

The smart narrative visualization helps you quickly summarize visuals and reports. It provides relevant innovative insights that you can customize

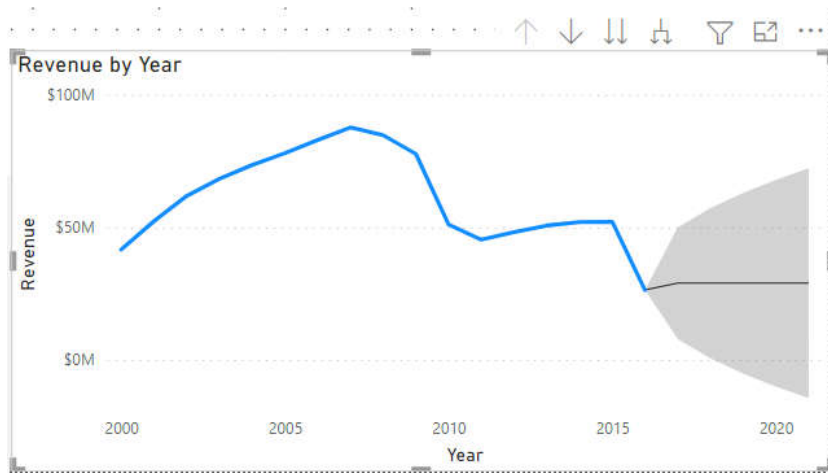
1. In the **Visualizations** pane, select the **Smart narrative** icon to automatically generate a summary.
2. You see a narrative that's based on all of the visuals on the page. For example, in the sample file, smart narratives can automatically generate a summary of the report's visuals that address revenue,



Power BI Desktop-Forecasting

Forecast next 5 years revenue.

1. Click new page report
2. From Visualizations section select Line Chart .
3. From the Data section expand Sales table and drag and drop Revenue to value.
4. From the Data section expand Date table and drag and drop Date to Axis
5. With the Line chart selected, from the Visualization click on the Analytics icon.
6. Expand Forecast and select on.
7. Set the Forecast length and Confidence Interval



Forecast length
5
Years

Ignore the last
0 Years

Confidence interval
95%

Seasonality
Auto Points

Apply

Axis

Date

Year

Quarter

Month

Day

Legend

Add data fields here

Values

Revenue

Secondary values